# HP-200 Hydraulic Forming Press — Operations Manual

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## Section 1: Machine Overview & Operating Specifications

The HP-200 is a 200-ton industrial hydraulic forming press engineered for precision metal stamping, deep drawing, and composite compression molding. It is equipped with a closed-loop proportional hydraulic system and digital programmable logic controller (PLC) architecture.

### Operating Limits and Pressure Specifications:

The maximum allowable operating pressure is 250 bar (3,625 psi). Under no operational circumstance or experimental tooling should the press be configured or operated above 250 bar under any load. Normal operating hydraulic pressure is set between 180 bar and 220 bar depending on forming tonnage demands.

Hydraulic Fluid Specifications: The reservoir capacity is 120 liters. Use only approved anti-wear ISO VG 46 hydraulic mineral oil. Hydraulic fluid operating temperature must remain between 35°C and 55°C. The maximum allowable oil temperature limit is 65°C, above which thermal degradation occurs.

Platen and Tonnage Limits: Maximum press capacity is 2,000 kN (200 metric tons). Stroke length is 450 mm. Daylight opening is 750 mm. Fast approach speed is 180 mm/s, pressing speed is 8-20 mm/s, and return speed is 160 mm/s.

## Section 2: Alarm Diagnostics & Error Codes

When the control system detects an operational fault, an audible beacon sounds and the diagnostic touch screen displays an alphanumeric error code. Refer to the diagnostic matrix below for required corrective actions.

| Alarm Code | Fault Description | Root Cause & Remedial Action |
|---|---|---|
| E17 | Main Ram Proportional Valve Command Signal Fault | Loss of command or position feedback deviation between the PLC axis controller a |
| E21 | Low Hydraulic Oil Level | The reservoir level float switch has tripped. Action: Immediately suspend automatic cyc |
| E04 | High Hydraulic Oil Temperature (>65°C) | Hydraulic bulk temperature sensor exceeded 65°C threshold. Action: Stop machine cyc |
| E09 | Safety Light Curtain Obstruction / Muting Failure | Optical beam interruption during down-stroke. Action: Clear foreign objects from light c |

## Section 3: Standard Operating & Safety Procedures

Shift Pre-Start Check: 1) Ensure main disconnect is locked ON; 2) Check that emergency stop buttons (platen front, rear, and pendant) are released; 3) Verify hydraulic fluid level is between MIN and MAX marks on the reservoir sight glass; 4) Check pressure gauge reads 0 bar before motor start.

Operating Cycle Modes: The HP-200 supports Inching (Manual Set-up), Single Cycle Semi-Automatic, and Continuous Automatic mode. Inching mode requires two-hand concurrent depression of dual optical palm buttons. Releasing either palm button within the down-stroke halts ram motion within 40 milliseconds.

Lockout / Tagout (LOTO) Regulations: Before any operator enters the die space for tool adjustment or cleaning, the mechanical ram safety restraint block must be swung into the vertical position beneath the upper platen, and the electrical lockout padlock applied to breaker CB1.